

Abstract

INPUT

image 

Label Class: Waterbird

Textspan An image of a Psychologist

FEATURES & WEIGHTS PREPROCESSING

Attention Heads

MLP

Classifier

Dataset Classifier

Textspan's embedding

HEAD ANALYSIS AND SELECTION

Selected heads:

Layer 18, Head 1: Urban intersection

Layer 18, Head 5: A wizard's hat

Layer 18, Head 7: Reflective pond scene

Layer 18, Head 8: An image of Andorra

Layer 18, Head 11: Image with a whirlpool in the ocean

Layer 18, Head 12: Photo taken in the Hawaiian volcanoes

ABLATION AND ANALYSIS

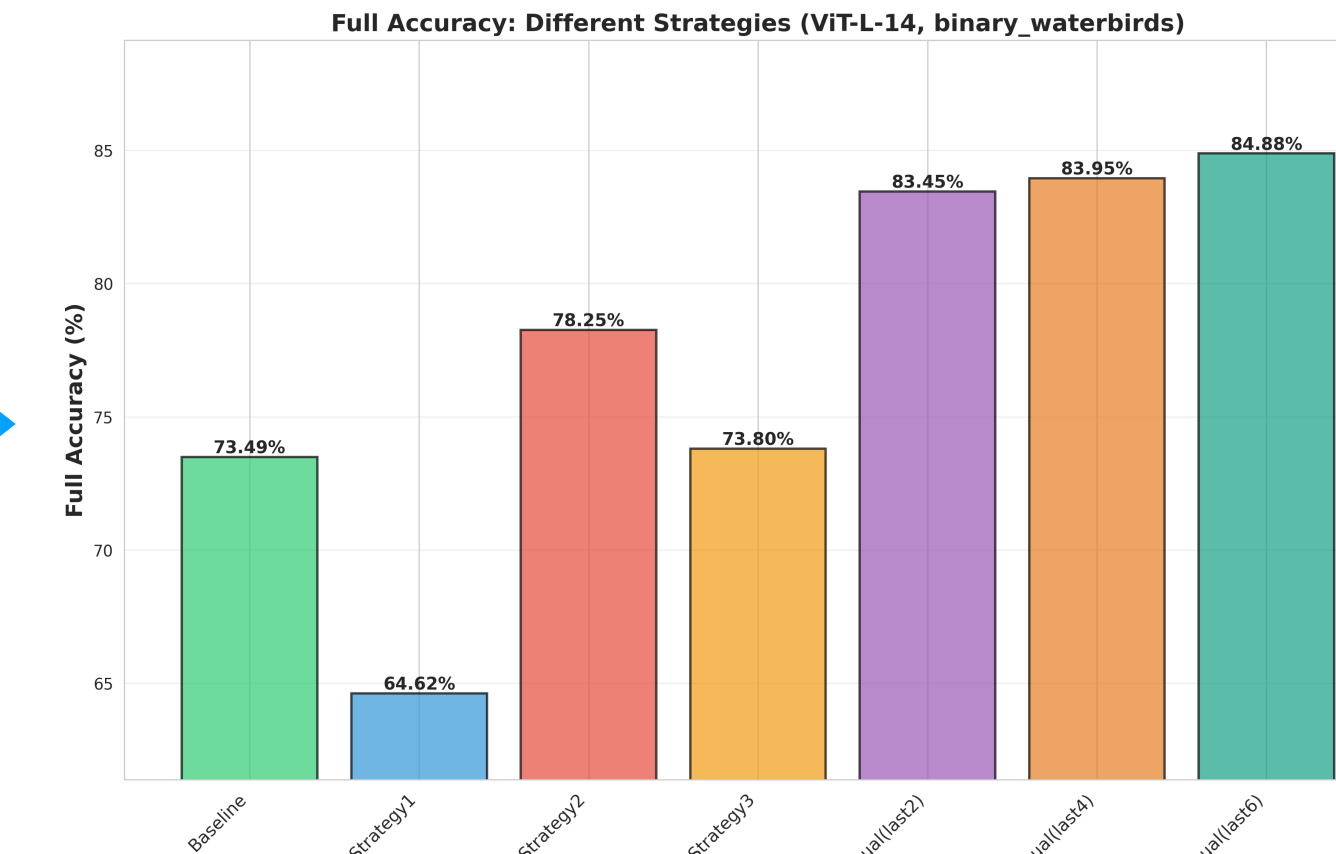


Figure 1: Pipeline from model training to experimental evaluation.

Modern vision-language models achieve high accuracy but often rely on spurious correlations. Instead of learning causal, bird-specific cues, they can latch onto background patterns that happen to co-occur with certain classes in training. As a result, these models might fail on minority subgroups, where those correlations are absent.

Problem: Lack of robustness and poor worst-group generalization.

Our approach: Identify and ablate attention heads that consistently focus on non-bird, background features.

Goal: Improve worst-group accuracy while persevere overall performance.

Methodology

Base model: OpenAI CLIP with Vision Transformer (ViT-L-14).

Pre-training: LAION-2B dataset (2B image-text pairs).

Architecture details: 24 transformer layers and 16 attention heads per layer.

We will focus on the last two layers since they are the closest to the final predictions. For each attention head, compute how its outputs project onto different semantic concepts. Generate attention maps showing what visual regions each head focuses on

Metrics:

Worst-group accuracy: Performance on minority subgroups.

Overall accuracy: Average across all test subgroups.

We denote each subgroup as (bird, scene) using:

(0, 0): Landbird on land scene (0, 1): Landbird on water scene

(1, 0): Waterbird on land scene (1, 1): Waterbird on water scene

Dataset

200-class bird species dataset. We use the dataset for a binary classification of waterbirds (e.g., seagulls, ducks) and landlords (e.g., sparrows, robins). The dataset has two backgrounds of water scenes and land scenes. The background of the images in the dataset do not have any clear correlation with the bird species

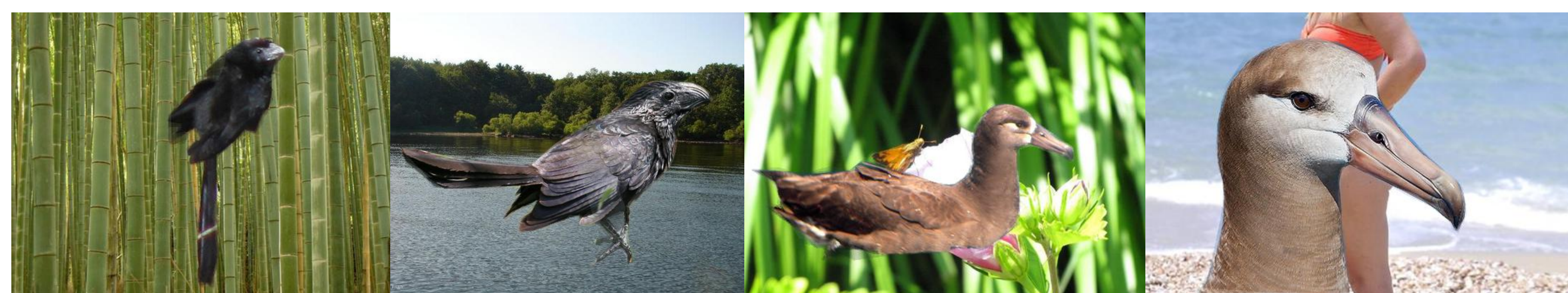


Figure 2: Four combinations of bird type and scene—landbird-land, landbird-water, waterbird-land, waterbird-water—shown left to right

Current Results

TextSpan of selected heads

Layer 18, Head 1: Urban intersection / Layer 18, Head 5: A wizard's hat / Layer 18, Head 7: Reflective pond scene / Layer 18, Head 8: An image of Andorra / Layer 18, Head 11: Image with a whirlpool in the ocean / Layer 18, Head 12: Photo taken in the Hawaiian volcanoes ...

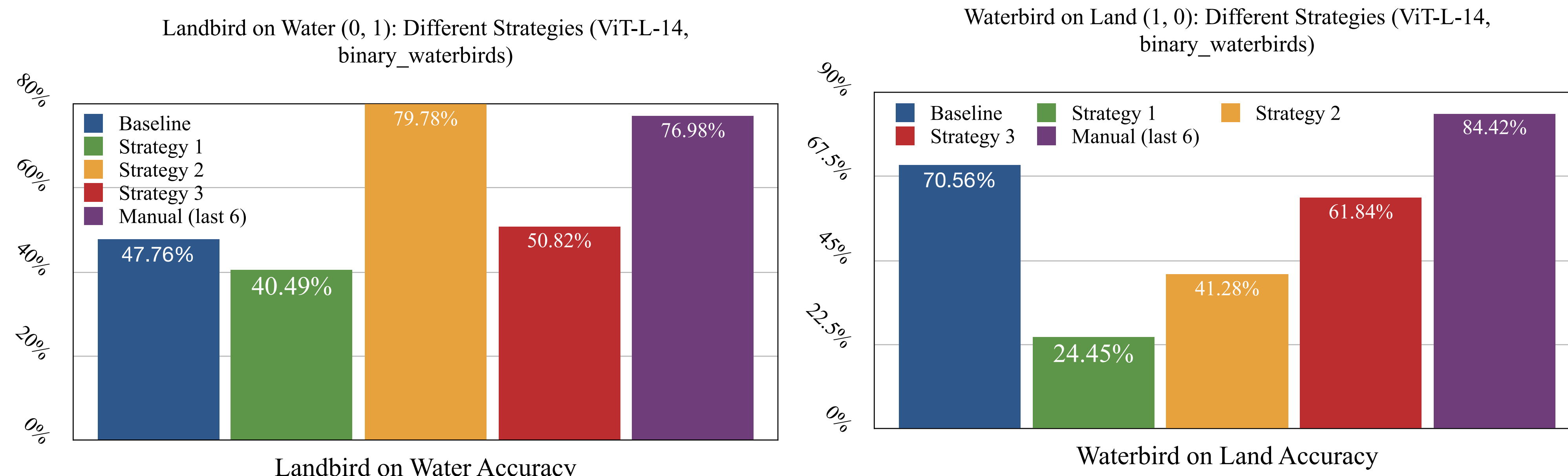


Figure 3: Compared to baseline model, we gained significant accuracy improvement using Strategy 2 and Strategy 4 for spurious subgroup landbird on water

Figure 4: Compared to baseline model, we could only gain significant accuracy improvement using Strategy 4 for spurious subgroup waterbird on land

Metric	Baseline	Strategy 1	Strategy 2	Strategy 3	Manual (last 2)	Manual (last 4)	Manual (last 6)
(0, 0)	94.41%	95.88% (+1.46)	95.30% (+0.89)	95.12% (+0.71)	92.24% (-2.17)	92.02% (-2.39)	91.53% (-2.88)
(0, 1)	47.76%	40.49% (-7.27)	79.78% (+32.02)	50.82% (+3.06)	72.55% (+24.79)	74.15% (+26.39)	76.98% (+29.22)
(1, 0)	70.56%	24.45% (-46.11)	41.28% (-29.28)	61.84% (-8.72)	83.33% (+12.77)	83.49% (+12.93)	84.42% (+13.86)
(1, 1)	93.30%	79.75% (-13.55)	50.00% (-43.30)	91.59% (-1.71)	90.97% (-2.34)	90.50% (-2.80)	89.72% (-3.58)
Full	73.49%	64.62% (-8.87)	78.25% (+4.76)	73.80% (+0.31)	83.45% (+9.96)	83.95% (+10.46)	84.88% (+11.39)

Table 1: Strategy 2, 3, and 4 all obtained higher overall accuracy score than baseline model. Strategy 3 and Strategy 4 obtained higher worst-group accuracy score compared to baseline model

Metric	Heads Only	+Early (keep 2)	+Early (keep 4)	+MLP	+Both (keep 2)	+Both (keep 4)
(0, 0)	95.30% (+0.89)	91.31% (-3.1)	91.84% (-2.57)	98.89% (+4.48)	98.89% (+4.48)	98.76% (+4.35)
(0, 1)	79.78% (+32.02)	92.37% (+44.61)	92.02% (+44.26)	75.57% (+27.80)	92.11% (+44.35)	90.86% (+43.10)
(1, 0)	41.28% (-29.28)	43.15% (-27.41)	45.33% (-25.23)	38.63% (-31.93)	35.36% (-35.20)	37.69% (-32.87)
(1, 1)	50.00% (-43.30)	18.54% (-74.77)	22.74% (-70.56)	78.50% (-14.80)	50.93% (-42.37)	57.63% (-35.67)
Full	78.25% (+4.76)	78.32% (+4.83)	79.10% (+5.61)	80.88% (+7.39)	83.90% (+10.41)	84.36% (+10.87)

Table 2: Accuracy of different ablation scopes with Strategy 2. Each ablation scope increased overall accuracy while reducing worst-group accuracy.

Ablation Strategies

Strategy 1: Largest Cluster Removal. The dominant attention pattern may encode the spurious correlation. Expected result: Reduce spurious reliance if the largest cluster focuses on backgrounds.

Strategy 2: Background Ablation. We target the attention heads that cause the wrong classification while preserving the attention heads focusing on bird features. Expected results: Improved worst-group accuracy with minimal overall accuracy loss.

Strategy 3: Representative Selection. Maintain diverse attention patterns while reducing redundancy. Expected results: More balanced performance across groups, potential efficiency gains.

Strategy 4: User-Guided Selection. Domain expert knowledge can identify optimal ablation targets Expected result: Improve worst-group accuracy while lowering overall performance because of the bias away from the objective function.

Further Discussion

To prevent multiple attention heads from collapsing onto the same spurious background feature, we can incorporate a diversity loss term.

Let X be the matrix of head outputs z_i . We compute the cosine-similarity matrix $S_{ij} = \frac{z_i \cdot z_j}{\|z_i\| \|z_j\|}$. Zero-out the diagonal,

apply ReLU to keep only positive correlations, and take the Frobenius norm. This yields a penalty that increases when heads become redundant. $L_{\text{diversity}} = \|S - I\|_F$

New objective function: $L_{\text{total}} = L_{\text{classification}} + \lambda L_{\text{diversity}}$

This regularizer forces late-layer heads to encode distinct visual features instead of converging on the same background-based shortcuts. As a result, the model should exhibit lower head redundancy and improved worst-group accuracy.

Reference

[1]Y. Gandelsman, A. Efros, and J. Steinhardt, "Published as a conference paper at ICLR 2024 INTERPRETING CLIP'S IMAGE REPRESENTATION VIA TEXT-BASED DECOMPOSITION." Accessed: Apr. 30, 2024. [Online]. Available: <https://arxiv.org/pdf/2310.05916>